Computational Methods for Macroeconomics

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Assignment 1

A household consumes and supplies labor in the following way:

$$\begin{aligned} R(a,z) &= \max\{W(a,z), N(a,z)\} \\ W(a,z) &= \max_{c,n,a'} \quad \log(c) - \eta \frac{1}{1+\frac{1}{\chi}} n^{1+\frac{1}{\chi}} + \beta v(a') \\ &\quad \text{s.t.} \quad c + a' = zw(1-\tau)n + a(1 + r(1-\tau)) + T \\ N(a,z) &= \max_{c,a'} \quad \log(c) + \beta v(a') \\ &\quad \text{s.t.} \quad c + a' = b + a(1 + r(1-\tau)) + T \end{aligned}$$

where $v(a') = \log(a')$. The idiosyncratic productivity z satisfies $\log(z) \sim_{iid} N(0, \sigma_z^2)$. The production sector operates as follows:

$$\max_{L} AK^{\alpha}L^{1-\alpha} - wL - rK$$

where $Y = AK^{\alpha}L^{1-\alpha}$ is the output, and the capital stock is exogenously given by the households' wealth, $K = \int a d\Phi = 1$. The government budget balance condition is as follows:

$$\underbrace{\int (wzn + ar)\tau d\Phi}_{\text{Revenue}} = \underbrace{\int_{\{\mathcal{NW}\}} b d\Phi + T}_{\text{Spending}}$$

Wage is determined in the following competitive effective labor market:

$$L^* = \int z n^*(z) d\Phi$$

1. Label and interpret the model ingredients properly.
2. Characterize the individual labor supply curve.
3. Characterize the aggregate labor supply curve.
4. Characterize the aggregate labor demand curve.

5. Suppose the following parameter levels:

$$a = 1, \quad \alpha = 0.3, \quad \tau = 0.15, \quad A = 1, \quad r = 0.04, \quad \beta = 0.96$$

   Define and characterize the stationary recursive competitive equilibrium.

6. (Method of simulated moments) Estimate the parameters $(\eta, b, \chi, \sigma_z)$ to match the following hypothetical moments in general equilibrium, where $\mathcal{W} = \{(a, z) : z \geq z^*(a)\}$ is the set of working households:

   - Average working hours among workers: $\int_{\mathcal{W}} n \, d\Phi / \int_{\mathcal{W}} d\Phi = 0.33$.
   - Unemployment rate: $\int_{z<z^*(a)} d\Phi = 0.06$ (Assume unemployed = non-working).
   - Unemployment benefit: $b / \left(\int_{\mathcal{W}} wzn \, d\Phi / \int_{\mathcal{W}} d\Phi \right) = 0.25$, i.e., 25% of the average wage income of workers.
   - STD/Mean ratio of wage income among workers: $std(wzn \,|\, \mathcal{W}) / mean(wzn \,|\, \mathcal{W}) = 0.70$.

7. Visualize the aggregate supply and demand curves in the labor market.

8. Visualize the comparative statics of the wage with respect to the change in A (horizontal axis: A, vertical: w).

9. Counterfactual exercise: How does an increase in the tax rate from 0.15 to 0.16 affect the total income inequality? (including non-working households). Check the various inequality measures.

10. Counterfactual exercise: Suppose η increases instantaneously by 20% (Say, Covid-19). How does it affect the total income inequality? (including non-working households). Check the various inequality measures.